

Highlights

Co-firing copper in a cordierite-type glass-ceramic printed by multi-material vat photopolymerisation: a model-based process design

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Abstract

Keywords: Co-firing, Glass-ceramic, Copper metallisation, Vat photopolymerisation, Binder removal, Sintering mismatch

1. Introduction

Packages for millimetre-wave front ends, antenna-in-package modules and wide-bandgap power devices now have to route conductors in three dimensions through a low-loss dielectric, and to do it close to the die [1, 2, 3, 4]. Co-fired ceramics answer part of this demand. Low-temperature co-fired ceramic (LTCC) and glass-ceramic substrates combine low dielectric loss, hermeticity and a thermal expansion close to that of silicon [5, 6]. They are, however, built from stacked tapes: conductors are printed in planes, interconnections run vertically through punched vias, and every change of geometry requires new tooling [7, 5]. The Heterogeneous Integration Roadmap lists additive manufacturing of ceramic packages, including lithography-based ceramic manufacturing, as one way past these limits [8].

The conductor of choice in fired packages has long been silver, because it can be co-fired in air. Silver migrates under bias and humidity and diffuses into the glass when fired in oxidising conditions [9, 10, 11]. Copper conducts almost as well, costs less and is far less prone to migration, but it must be fired in an atmosphere that keeps it metallic. In such an atmosphere the organic binder no longer burns off cleanly: part of it is left as char, and char trapped in a glass that has already sealed its pores degrades dielectric loss and can bloat the part [12]. IBM solved this for its glass-ceramic/copper multilayer substrates with a steam-based co-sintering process that gave a dense glass-ceramic with high-conductivity copper [13, 6]. That solution was developed for laminated tapes and its operating window was found empirically.

Vat photopolymerisation of ceramic slurries, in particular lithography-based ceramic manufacturing (LCM), produces dense ceramics with features of a few tens of micrometres [14, 15, 16]. It has already been used for monolithic microwave filters and antenna substrates [17, 18, 19], and photocurable LTCC suspensions have been formulated for it [20]. In these parts the metallisation is still added afterwards, by plating or by printing on the fired ceramic [17, 21], or the conductors are jetted into ceramic layers by a separate route [22, 23, 24, 25].

Multi-material vat photopolymerisation removes that restriction. Printers such as the CeraFab Multi 2M30 switch between slurries within a layer [26, 27, 28]. Co-sintered ceramic–ceramic parts have been demonstrated [29], as has a zirconia part with titanium-oxide conductors co-fired in a reducing atmosphere [30]. Copper slurries for vat photopolymerisation have been developed separately [31, 32, 33, 34]. In principle, then, a glass-ceramic body and its copper wiring can be printed as one green part. We found no study of whether such a part can be fired.

Four requirements make that firing difficult, and they are coupled. First, the char from both slurries must be removed before the glass seals its pores, in an atmosphere that does not oxidise the copper. Second, the two materials must shrink at compatible times and by compatible amounts; otherwise the part cambers, delaminates or cracks around its conductors [35, 36, 37]. Third, the glass must densify before it crystallises, because crystals stop viscous flow. Fourth, a thermal-expansion mismatch of about 14 ppm K^{-1} between copper and a cordierite-type glass-ceramic must be accommodated on cooling. In tape-based LTCC the second requirement is met by trial, most often by adding ceramic powder to the metal paste [38] or by adjusting the heating rate [39]. Continuum models of co-sintering reproduce camber and mismatch stresses in ceramic and metal–ceramic bilayers [40, 41, 42, 43, 44, 45], but none includes the chemistry of binder removal, and none treats copper with a crystallising glass. The first and third requirements are therefore usually handled separately from the second, although in practice they compete for the same temperature range.

This paper sets out a model-based design of that firing for a copper conductor printed inside a cordierite-type glass-ceramic body. We extend a verified slab model of binder removal and sintering, originally developed for printed copper, to a crystallising glass-ceramic, and couple the two materials through the viscous analogue of laminate theory and a three-dimensional finite-element model built on the printed voxel grid. The model is verified against closed-form solutions and parameterised from published data, with every input tagged by its provenance. With it we (i) map the atmosphere and temperature window in which carbon can be removed from the glass before its pores close while copper stays metallic; (ii) show that fine copper and a fragile glass densify in different temperature ranges and with

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different curve shapes, and quantify the consequences for camber and conductor damage; (iii) design the copper paste and the firing programme together to bring the two into step; and (iv) identify the few material parameters that decide the outcome, and should therefore be measured first.

2. Model formulation

2.1. Structure of the model

The co-fired part is represented at two levels (Fig. 1b). At the first level, each material is followed through the firing programme by a one-dimensional slab model that resolves heat transfer, binder decomposition, char formation and removal, pore closure and densification through the thickness of its critical section. The glass-ceramic body and the copper conductor are simulated separately but share the furnace set point and the inlet gas, so both see the same thermal and chemical history. At the second level, the two densification histories are coupled mechanically: analytically, through the viscous analogue of laminate theory, and numerically, through a three-dimensional finite-element model in which both materials occupy the voxels of the printed part. Both levels use the same constitutive law for the porous body, so a free sample in the finite-element model reproduces the slab model exactly.

The copper slab model was developed and verified for single-material copper parts printed from a lithography-based slurry; its chemistry and its calibration against published copper data are summarised in Appendix A. The additions made here are the crystallising glass-ceramic, the copper paste with an inert filler, the steam burnout atmosphere, and the co-firing mechanics.

2.2. Binder removal and residual carbon

The photocured binder is lumped into three pseudo-components i (a non-reactive solvent, a crosslinked network and a char-forming backbone), each decomposing by first-order kinetics,

$$\frac{dB_i}{dt} = -k_i(T) B_i, \quad k_i(T) = \frac{E_i \beta_{\text{ref}}}{RT_{p,i}^2} \exp\left[-\frac{E_i}{R} \left(\frac{1}{T} - \frac{1}{T_{p,i}}\right)\right], \quad (1)$$

where $T_{p,i}$ is the temperature of the corresponding peak in a thermogravimetric trace at $\beta_{\text{ref}} = 10 \text{ K min}^{-1}$. Writing the rate constant in terms of the peak temperature keeps the two fitted quantities nearly independent and lets them be read directly from a single thermogravimetric curve. A fraction χ of the network and backbone mass remains as char. In a steam-bearing atmosphere the char is gasified by $\text{C} + \text{H}_2\text{O} \rightarrow \text{CO} + \text{H}_2$ at the rate

$$r_g = a(\varepsilon_o) (1 - f_{\text{cl}}) k_g(T) \frac{c}{M_C} \left(\frac{x_{\text{H}_2\text{O}}}{x_{\text{ref}}}\right)^{1/2} \frac{1 + K_H x_{\text{H}_2}^{\text{ref}}}{1 + K_H x_{\text{H}_2}} \left(1 - \frac{Q}{K_{\text{st}}(T)}\right)_+, \quad (2)$$

in which c is the char concentration, $a(\varepsilon_o)$ the fraction of pore surface reachable through open porosity ε_o , f_{cl} the closed-pore fraction, K_H an inhibition constant for hydrogen adsorption, $Q = x_{\text{CO}} x_{\text{H}_2} / x_{\text{H}_2\text{O}}$ the reaction quotient and K_{st} the equilibrium

constant from NIST-JANAF data [46]. Char in closed pores is not gasified: once the pores seal, whatever carbon remains stays in the part. The rate constant k_g is parameterised by a half-life at 800°C in the reference atmosphere and an activation energy; no measured value exists for char on either material, so both enter the sensitivity analysis. A Thiele-modulus estimate ($k_g L^2 / D_{\text{eff}} \sim 10^{-4}$ for $L = 2 \text{ mm}$) shows that gasification is kinetically controlled, so steam access is represented by the open-pore accessibility $a(\varepsilon_o)$ rather than by a diffusion equation.

Pyrolysis gas raises the pore pressure. The overpressure is computed from the compressible Darcy equation with Kozeny–Carman and Knudsen contributions to the permeability and a polymer-permeation term that carries the flux before the pore network percolates; it is compared with the green or brown strength through a safety factor. The same slab tracks the through-thickness temperature difference and converts it into a thermal-stress index.

2.3. Densification

Both materials densify according to the Skorohod–Olevsky viscous sintering model [47]. For a porous body of porosity θ and matrix shear viscosity η , the stress is

$$\sigma = 2\eta [\varphi(\theta) \dot{\varepsilon}' + \psi(\theta) \text{tr}(\dot{\varepsilon}) \mathbf{I}] + P_L(\theta) \mathbf{I}, \quad (3)$$

with $\varphi = (1 - \theta)^2$, $\psi = \frac{2}{3}(1 - \theta)^3 / \theta$ and sintering stress $P_L = 3\gamma(1 - \theta)^2 / r$, where γ is the surface energy and r half the Sauter mean particle diameter. A free body densifies at $\text{tr}(\dot{\varepsilon}) = -(P_L - \Delta p) / (2\eta\psi)$, where Δp is the pressure of insoluble gas trapped in closed pores. The two materials differ only in η , γ and r .

Glass-ceramic. The residual melt is Newtonian. Its viscosity follows the MYEGA equation [48],

$$\log_{10} \eta_g = \log_{10} \eta_\infty + (12 - \log_{10} \eta_\infty) \frac{T_g}{T} \exp\left[\left(\frac{m}{12 - \log_{10} \eta_\infty} - 1\right) \left(\frac{T_g}{T} - 1\right)\right] \quad (4)$$

with $\log_{10} \eta_\infty = -3 \text{ (Pa s)}$, glass transition T_g (defined at 10^{12} Pa s) and fragility index m . Crystallisation during firing follows non-isothermal Johnson–Mehl–Avrami–Kolmogorov kinetics [49] written in terms of the extended-volume integral,

$$\xi = k_c(T) = k_{c,p} \exp\left[-\frac{E_c}{R} \left(\frac{1}{T} - \frac{1}{T_{p,c}}\right)\right], \quad X = 1 - \exp(-\xi^n), \quad (5)$$

which avoids the singularity of the rate form at $X = 0$ for $n > 1$. The constant $k_{c,p}$ is fixed numerically so that the simulated crystallisation exotherm at 10 K min^{-1} peaks exactly at the measured temperature $T_{p,c}$; Kissinger's closed form [50] is exact only for $n = 1$ and would misplace the peak by about 1 K for $n = 2$. Crystals stiffen the melt as rigid inclusions,

$$\eta = \eta_g(T) \left(1 - \frac{X}{X_m}\right)^{-[\eta]X_m} \left(1 - \frac{f_c}{0.64}\right)^{-[\eta]0.64}, \quad (6)$$

the Krieger–Dougherty law [51] with intrinsic viscosity $[\eta] = 2.5$ and maximum packing X_m . The second factor treats residual char particles, volume fraction f_c , the same way. Once X approaches X_m the body stops flowing, which is how crystallisation arrests densification.

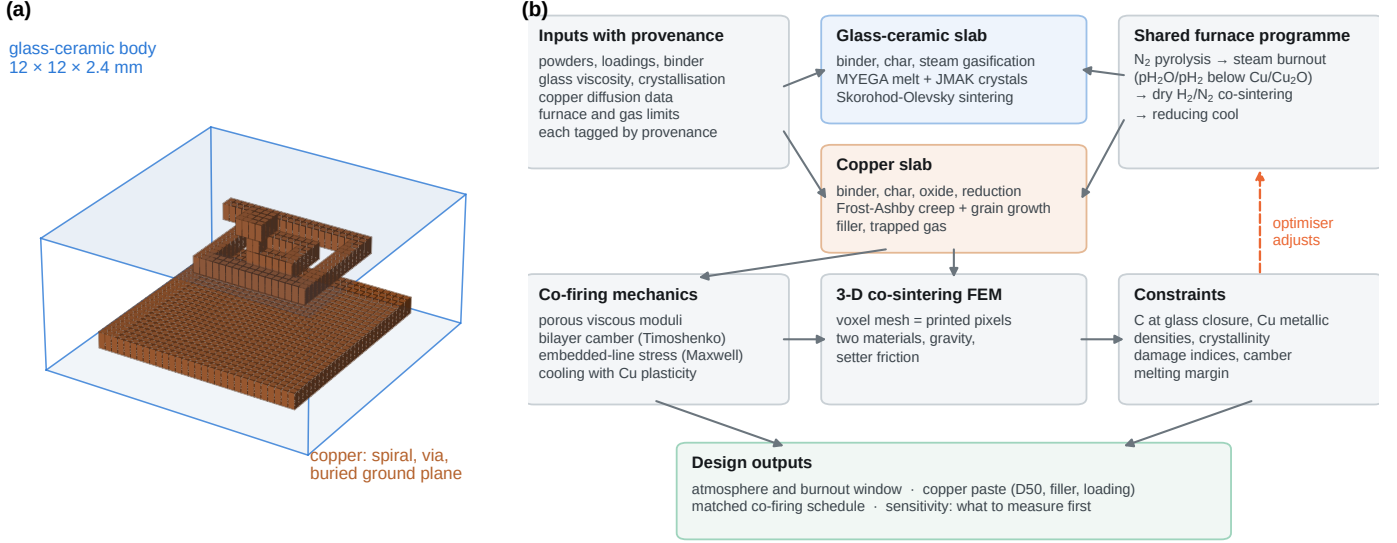


Figure 1: (a) Test vehicle: a 12 × 12 × 2.4 mm glass-ceramic body with a buried copper ground plane, a square spiral and a via to a top pad, printed as one part. (b) Structure of the model. Each material is followed by its own slab model under a shared furnace programme; the two histories are coupled mechanically, analytically and in three dimensions, and the programme and paste are searched against constraints.

Copper. The copper skeleton creeps by diffusion. Its matrix viscosity combines lattice and grain-boundary diffusion [52],

$$\eta_{\text{Cu}} = f_{\eta} \frac{k_B T G^2}{42 \Omega (D_v + \pi \delta D_b / G)} \left(1 + \frac{C^2}{C_{\text{inh}}^2} \right) \left(1 - \frac{f}{0.64} \right)^{-1.6}, \quad (7)$$

where G is the grain size, Ω the atomic volume and f_{η} a calibration factor fitted to published copper densification data (Appendix A). The carbon term C (in ppm) describes the retarding effect of residual carbon on copper, and the last factor accounts for an inert ceramic filler of volume fraction f added to the conductor paste, a standard way to slow a metal paste down to the pace of its ceramic [38]. Grain growth is parabolic, driven by grain-boundary diffusion and pinned by porosity. The filler also lowers the conductivity; together with porosity,

$$\frac{\sigma}{\sigma_0} = (1 - \theta)^{3/2} \frac{1 - f}{1 + f/2}, \quad (8)$$

the second factor being the Maxwell Garnett result for insulating inclusions [53].

2.4. Atmosphere and the metallic state of copper

Steam removes char but oxidises copper once the water-to-hydrogen ratio exceeds the equilibrium of $\text{Cu}_2\text{O} + \text{H}_2 \rightleftharpoons 2\text{Cu} + \text{H}_2\text{O}$,

$$\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} < K_{\text{Cu}}(T) = \exp\left(-\frac{\Delta G^\circ(T)}{RT}\right). \quad (9)$$

During the burnout segment the inlet hydrogen is therefore tied to the steam content, $x_{\text{H}_2} = x_{\text{H}_2\text{O}}/(s K_{\text{Cu}}(T_B))$, with a safety factor $s = 0.2$. The copper stays metallic by construction, and the steam is as oxidising towards carbon as the metal allows. This is the principle of the steam co-sintering developed for glass-ceramic/copper substrates [13, 6]; here it is applied in

a model that says how long, at what temperature and in how much steam. Carbon gasification itself is thermodynamically unconstrained over the whole window (Section 4.2); its limits are kinetic.

2.5. Mechanics of co-firing

Porous viscosities. Eq. (3) describes an isotropic, linear viscous solid with shear viscosity $G_v = \eta\varphi$ and bulk viscosity $K_v = 2\eta\psi$, hence uniaxial viscosity $E_v = 9K_v G_v / (3K_v + G_v)$, viscous Poisson ratio $\nu_v = (3K_v - 2G_v) / [2(3K_v + G_v)]$ and biaxial viscosity $B = E_v / (1 - \nu_v)$. The free linear sintering rate is $\dot{\epsilon}_f = -P_L / (6\eta\psi)$. All three quantities follow from the slab histories of T , θ , G or ξ , and carbon.

Camber. By the elastic-viscous correspondence principle, a bilayer of two linear viscous layers bends at a curvature rate given by Timoshenko's bimetal solution [54], with elastic moduli replaced by biaxial viscosities and thermal strain by the difference in free sintering rates [55, 37, 40]:

$$\dot{\kappa} = \frac{6(1 + \lambda)^2 (\dot{\epsilon}_{f,2} - \dot{\epsilon}_{f,1})}{h \left[3(1 + \lambda)^2 + (1 + \lambda\mu) \left(\lambda^2 + \frac{1}{\lambda\mu} \right) \right]}, \quad \lambda = \frac{h_1}{h_2}, \quad \mu = \frac{B_1}{B_2}. \quad (10)$$

Layer 1 is copper and layer 2 the glass-ceramic; positive curvature is concave towards the copper. Eq. (10) is integrated over the firing with the layer thicknesses updated by their own shrinkage. The reference stack is a 0.1 mm copper plane on 1 mm of glass-ceramic.

Embedded conductors. A conductor line buried in the body cannot curl; its misfit is taken up by stress. We represent line

and surrounding body as two Maxwell elements in series, each with a porous elastic modulus and a biaxial viscosity:

$$\dot{\epsilon}_{f,GC} - \dot{\epsilon}_{f,Cu} = \dot{\sigma} \left(\frac{1}{E_{Cu}} + \frac{1}{E_{GC}} \right) + \sigma \left(\frac{1}{B_{Cu}} + \frac{1}{B_{GC}} \right). \quad (11)$$

A compliant line follows the body and carries B_{Cu} times the misfit rate; a stiff line forces the body to flow around it. Tension in the copper ($\sigma > 0$) opens voids and breaks the line; compression in the copper means the body is shrinking onto a stiff inclusion and is itself in hoop tension, the classic origin of radial cracks around conductors [35, 36]. Two damage indices compare these stresses with the strength of the weaker partner at that moment,

$$\Pi_{Cu} = \frac{\max(\sigma, 0)}{\sigma_{s,Cu}(\rho_{Cu})}, \quad \Pi_{GC} = \frac{\max(-\sigma, 0)}{\sigma_{s,GC}(\rho_{GC})}, \quad (12)$$

where the strengths σ_s grow from the binder-carried green strength through the neck-controlled brown strength. The series coupling carries an Eshelby factor of order unity [56] that we do not resolve, so the indices are used comparatively. The same coupling is resolved exactly in three dimensions below.

Three-dimensional co-sintering. The printed part is meshed voxel by voxel with trilinear hexahedra, so the mesh is the printed pixel grid. Each element carries its material, porosity and microstructural state. The velocity field satisfies $\nabla \cdot \sigma + \rho_b \mathbf{g} = \mathbf{0}$ with Eq. (3), gravity, and a setter plate beneath the part: zero normal velocity and regularised Coulomb friction ($\mu = 0.6$) iterated to convergence at each step. Volumetric terms use one-point quadrature, which avoids locking as $\theta \rightarrow 0$. The problem is assembled with scikit-fem [57] and solved by conjugate gradients preconditioned with smoothed-aggregation algebraic multigrid carrying the six rigid-body modes [58]. Porosity is advanced with the exact mass-conserving update $1 - \theta^{n+1} = (1 - \theta^n) \exp(-\text{tr } \dot{\epsilon} \Delta t)$. The temperature and carbon histories are taken from the two slab simulations.

Cooling. Below a stress-free temperature T_{sf} , where the residual glass stops relaxing, the thermal-expansion mismatch is carried elastically by the glass-ceramic and elastoplastically by the copper. The copper layer is treated as elastic-perfectly plastic with a temperature-dependent yield stress; its mean stress is capped at yield by a uniform plastic strain, and the curvature follows from Timoshenko’s elastic solution for the remaining elastic misfit.

2.6. Numerical solution, design search and sensitivity

The slab equations are integrated with a linearly implicit Rosenbrock method of orders 2(3) [59] on a geometrically graded mesh that is finest at the surface. A co-firing evaluation, two slab simulations and the mechanical post-processing, takes about 6 s. Designs were searched by differential evolution [60] over eleven variables: copper median particle size, filler fraction and solids loading; burnout temperature, time and steam fraction; temperature and duration of an intermediate matching hold; heating rate to the peak; peak temperature and hold. The objective was the largest free-strain mismatch between the two materials during firing, $\max_t |\epsilon_{f,GC} - \epsilon_{f,Cu}|$, which governs both

camber and conductor stress. It was minimised subject to hard constraints and to a cycle-time budget of 18 h, the ϵ -constraint form of the trade-off between cycle time and mismatch. The constraints were: carbon at glass pore closure below 100 ppm and in the copper below 200 ppm; relative densities of at least 0.97 for the glass-ceramic and 0.92 for the copper; a crystallised fraction of at least 0.8; a line damage index below 1; no trapped-gas pressure above 1.2 times the sintering stress; and a positive margin to the melting point of copper. Violations entered the objective as a penalty proportional to their size, so the search is steered back towards feasible designs. The population of 88 designs was evolved for 15 generations.

The influence of uncertain inputs was screened with Morris elementary effects [61]: 16 trajectories through a four-level grid over fourteen factors, 240 co-firing evaluations in total, around the optimised design. Twelve factors are the uncertain inputs with the ranges of Table 1; the other two are manufacturing tolerances on the copper paste, -25 to $+33\%$ in particle size and ± 0.05 in filler fraction. For each output the mean absolute elementary effect μ^* measures the overall influence of a factor and the standard deviation σ its non-linearity and interaction with other factors.

3. Parameters and calibration

3.1. Provenance of the inputs

No measurements were made for this study. Every input therefore comes from one of five sources, and Table 1 states which one applies. A value is labelled L when it was reported for the glass-ceramic of the IBM process or for copper itself; A when it was taken from an analogous material, usually a commercial copper-compatible LTCC; C when it was calibrated here against published process observations (Section 3.2); D when it follows from other entries of the table; and E when it is an engineering estimate. Estimated and analogue values that matter enter the sensitivity analysis of Section 4.10, with the ranges given in the table.

The reference glass is the MgO-rich cordierite-type glass used by IBM for its copper-compatible substrates (glass #12: 52.5 SiO₂, 22.0 MgO, 22.0 Al₂O₃, 1.5 P₂O₅, 0.5 B₂O₃, 1.5 ZrO₂, wt%) [62, 63]. It was chosen because it is the only copper-compatible glass-ceramic for which the firing process, the atmosphere and the reasons behind the chosen temperatures have been published in detail. Its viscosity has not been published in a form we could read, so it was calibrated.

3.2. Calibration of the glass against the IBM process

The patents that describe the IBM steam co-sintering process state, for glass #12, why the burnout hold sits where it does. The hold is placed at 785 ± 10 °C, between the annealing and softening points of the glass. At 750 °C carbon removal would take “prohibitively excessive amounts of time”; at 830 °C the glass pores close and trap water and binder residue [62]. A later patent from the same group reports that 33–50% of the total fired shrinkage occurs during the burnout step [78], and a third places the onset of crystallisation at about 900 °C, after the burnout

hold and before the crystallisation hold at about 960 °C [79]. We simulated the full worked schedule of Ref. [62] (N_2 to 200 °C, H_2/H_2O to 450 °C, 2.9 K min⁻¹ to the hold, 6 h at H_2/H_2O = 10^{-4} , 0.5 h in N_2 , 2.1 K min⁻¹ to 960 °C, 2 h) and used these statements as targets. The patent specifies the H_2/H_2O ratio but not the dilution; we assumed 50% steam in nitrogen.

Two of them fix the viscosity. Requiring that a 6 h hold at 780 °C produce 40% of the fired shrinkage with the pores still open, and that the pores close within 2 h at 830 °C, gives $\log_{10} \eta = 10.62$ at 780 °C and 9.37 at 830 °C. The MYEGA curve through these two points, with $\log_{10} \eta_\infty = -3$, has $T_g = 733.6$ °C and $m = 32.4$. The remaining observations were not used in the fit and serve as checks. Between 800 and 860 °C the calibrated viscosity falls by one decade per 43 K. Giess et al. found that the sintering of an IBM cordierite-type glass over the same range followed the viscosity of the glass, which fell by about one decade per 40 K [65]. The annealing point, taken as 10^{12} Pa s, lies at 734 °C and the dilatometric softening point ($10^{6.6}$ Pa s) at 980 °C, which brackets the 785 °C hold as the patent requires. Stoichiometric cordierite glass is one to four decades more viscous over 780–960 °C [80]. Glass #12 contains more MgO and less Al_2O_3 than cordierite, together with B_2O_3 and P_2O_5 , so a lower viscosity is plausible, but this comparison is not a test of the fit.

The crystallisation kinetics were treated in the same way. With the activation energy measured on an IBM-type glass ($E_c = 303$ kJ mol⁻¹ [67]) and $n = 2$, placing the onset of crystallisation (5% crystallised) at 900 °C in the worked schedule requires an exotherm peak at 1040 °C at 10 K min⁻¹. The simulated exotherm then spans 978–1066 °C, a width of about 90 K, which agrees with the 80–100 °C window that the IBM patent associates with the crystallisation exotherm of its glasses [63]. Differential thermal analysis of other MgO– Al_2O_3 – SiO_2 glasses places the α -cordierite peak at 1006–1032 °C [64, 69].

Table 2 summarises the result. The 785 °C hold removes the char and leaves 47% of the shrinkage to the burnout step, inside the 33–50% reported for the process, while the pores stay open. At 830 °C the body densifies completely during the hold and seals its pores. These two rows reproduce the calibration targets and are therefore not independent evidence. The 750 °C row is: it depends on the gasification kinetics, which were not fitted, and it leaves four orders of magnitude more carbon than the 785 °C hold. The model does not reproduce the absolute time scale of the patent’s statement. With the estimated gasification rate, 33 ppm of carbon remain after 6 h at 750 °C, which may or may not have been acceptable for IBM; no carbon specification was published. For all three holds the final body is 99% dense and fully crystallised, and its free linear shrinkage is 23.1%.

4. Results

4.1. Verification

The model was checked against closed-form solutions before it was used for design (Appendix B). Isothermal densification of the glass-ceramic follows the Skorohod–Olevsky solution to within 1.3×10^{-4} in porosity for 4 to 16 cells (Fig. 2a), and the

three-dimensional curvature rate of a free copper/glass-ceramic plate converges to the viscous Timoshenko value at second order, with an error of 0.05% at 0.15 mm voxels (Fig. 2b). The analytical camber of Eq. (10) and the finite-element model therefore describe the same mechanics, and the analytical form can be used inside the design search.

4.2. Atmosphere window for burnout

Fig. 3 places the burnout atmosphere in the plane of temperature and p_{H_2O}/p_{H_2} . Copper remains metallic below the orange line, which falls from $10^{6.2}$ at 500 °C to $10^{4.8}$ at 780 °C and $10^{4.1}$ at 1000 °C. Carbon gasification by steam is thermodynamically possible above the dashed line, which lies eight or more decades lower. The two constraints therefore leave a wide band of permissible atmospheres, and the choice within it is kinetic. At 780 °C, 30% steam with hydrogen at 20% of the copper limit ($p_{H_2O}/p_{H_2} = 1.2 \times 10^4$) gives a char half-life of 0.43 h. Forming gas (4% H_2) humidified through a bubbler at 20 °C, a common choice for copper, contains 2.3% water; its half-life at the same temperature is 3.1 h, seven times longer, because the steam partial pressure is low and hydrogen inhibits the reaction.

The operating point of the IBM process, $H_2/H_2O = 10^{-4}$ at 785 ± 10 °C [62], lies at 17% of the copper limit and almost coincides with the steam point in Fig. 3. The rule used here, hydrogen tied to steam so that the water-to-hydrogen ratio sits at a fixed fraction of the copper limit, therefore reproduces the atmosphere that was found empirically for tape-cast substrates. It also says how the ratio should change if the burnout temperature changes: by a factor of about 2.5 per 100 K, following $K_{Cu}(T)$.

4.3. Removing carbon before the glass seals

The char has to leave the glass before its pores close, and the glass carries far more of it than the copper does. After pyrolysis in nitrogen the glass-ceramic contains 22 500 ppm carbon relative to its solids, the copper 5 300 ppm. The difference follows from the densities: both slurries hold a similar volume of binder, but a unit volume of glass-ceramic solids weighs a third as much as the same volume of copper. Carbon on the glass is also the more harmful, since whatever is sealed in reacts with the water released during crystallisation and bloats the part [62].

Fig. 4 maps the carbon left in the glass-ceramic at the moment its pores close, as a function of burnout temperature and hold time. In steam (Fig. 4a) the 100 ppm contour runs from 16 h at 710 °C to 1.1 h at 830 °C, and the window closes abruptly at 850 °C, where the glass seals during the hold. The IBM hold of 6 h at 785 °C falls in the region with less than 10 ppm. In humidified forming gas (Fig. 4b) only the longest hold, 16 h at 790 °C, brings the carbon down to 350 ppm; every other combination leaves more than 1000 ppm. Residual carbon also slows the densification of the glass. Acting as rigid inclusions, the 0.3–2% of carbon retained in forming gas lowers the final density from 0.99 to 0.96–0.99. Steam is therefore not an option for this material pair but a requirement.

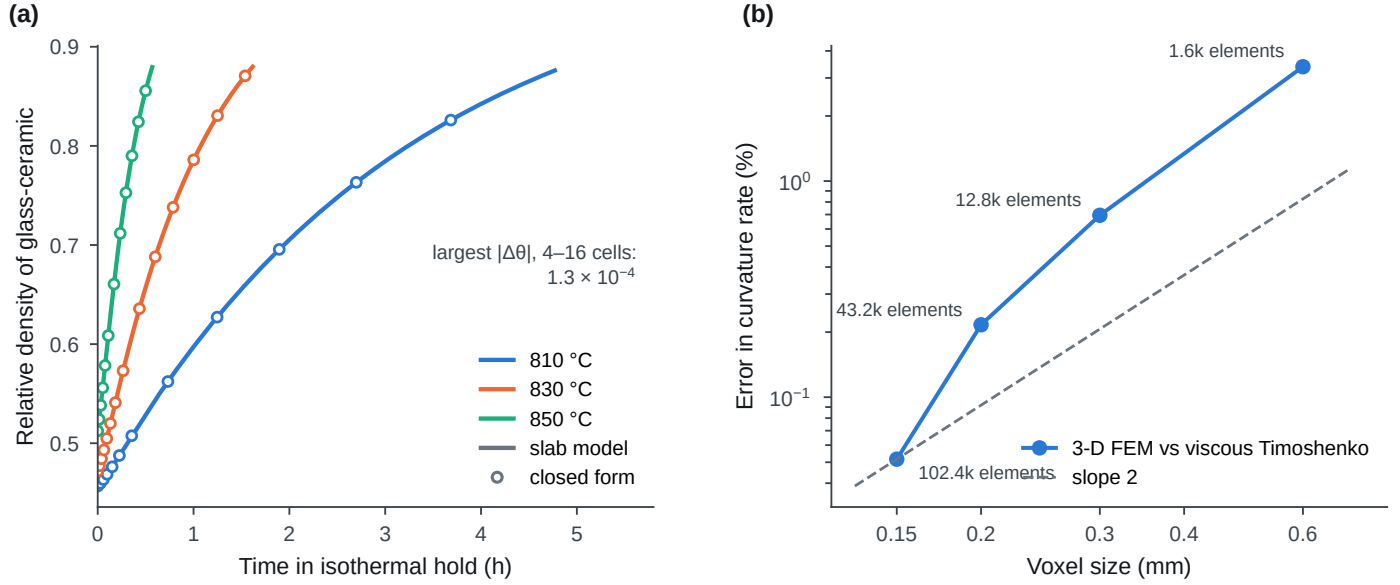


Figure 2: Verification. (a) Isothermal densification of the glass-ceramic slab at 810, 830 and 850 °C (lines, 8 cells) against the closed-form solution integrated along the simulated part temperature (circles); crystallisation and char were switched off. (b) Error of the three-dimensional curvature rate of a free 12 × 12 mm plate (0.6 mm copper on 1.8 mm glass-ceramic, 900 °C) against the viscous Timoshenko solution, Eq. (10), as the voxel size is refined.

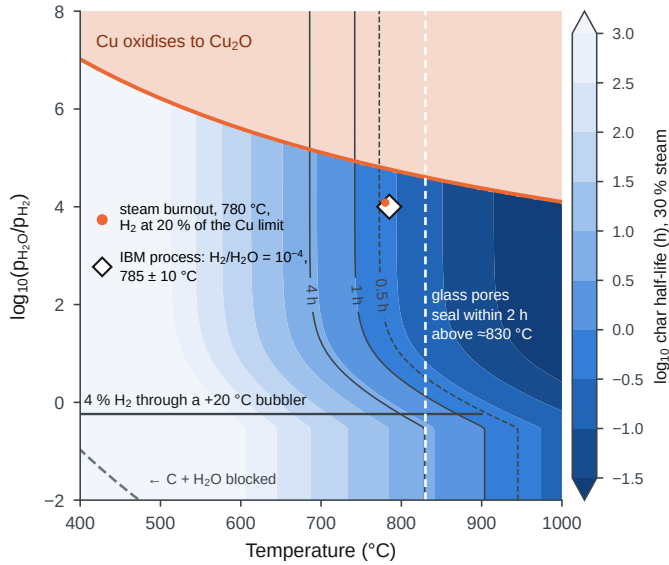


Figure 3: Burnout atmosphere window. Colour: half-life of char in 30% steam at the given $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$ (reference kinetics of Table 1). Copper oxidises above the orange line; carbon cannot be gasified to the left of the dashed line ($p_{\text{CO}} = 10^{-4}$ atm). Circle: steam burnout used in this work. Diamond: operating point of the IBM process [62]. The white line marks the temperature above which a 2 h hold seals the pores of the calibrated glass.

4.4. Two densification ranges

Fig. 5 compares the free sintering of the two materials after binder removal at 5 K min⁻¹ in dry hydrogen. The glass-ceramic shrinks by 23% between 850 and 933 °C (5 and 95% of its shrinkage) and then stops, because it crystallises: 93% of the glass has crystallised at 1060 °C. Fine copper densifies much earlier, and over a much wider range. A 3 μm powder shrinks between 454 and 702 °C; a 1 μm powder between 323 and

463 °C. Only for powders of 12 μm or coarser does the midpoint of densification fall inside the range of the glass, and these powders do not reach high density before 1060 °C: 83% for 12 μm and 60% for 20 μm .

The two curves also differ in shape. The glass densifies over a range of about 80 K, as a Newtonian melt whose viscosity falls steeply with temperature. Copper densifies over 140–350 K, because diffusional creep has a lower activation energy and grain growth slows it as it proceeds. Adding an inert filler to the paste delays and stretches the copper curve (Fig. 5b): 40% filler shifts the midpoint of 3 μm copper from 573 to 657 °C and widens the 5–95% range from 248 to 367 K. Neither particle size nor filler, on its own, gives copper a shrinkage curve that is as late and as steep as that of the glass. The matching has to come partly from the furnace programme.

4.5. Co-firing with an unmatched paste

The baseline design combines a 3 μm copper paste without filler with a programme that follows the IBM pattern: pyrolysis in nitrogen to 460 °C, a 4 h burnout at 780 °C in 30% steam, and sintering at 960 °C for 1 h in dry 4% H_2 (Fig. 6a). Chemically, the design works. The glass-ceramic retains 33 ppm of carbon when its pores close and reaches 99.1% density and full crystallinity; the copper is 97.6% dense, retains 81 ppm of carbon and conducts at 96% IACS.

Mechanically, it fails. As soon as steam begins to remove the char, the copper starts to sinter, while the glass is still rigid. At the end of the burnout hold the logarithmic shrinkage of the copper is 21% and that of the glass 7%; the free-strain mismatch peaks at 15.6%, and a free bilayer of 0.1 mm copper on 1 mm glass-ceramic would reach a curvature of 58 m⁻¹, concave towards the copper. The damage index of an embedded line reaches $\Pi_{\text{Cu}} = 1.6$ at 680 °C, during the ramp to the burnout

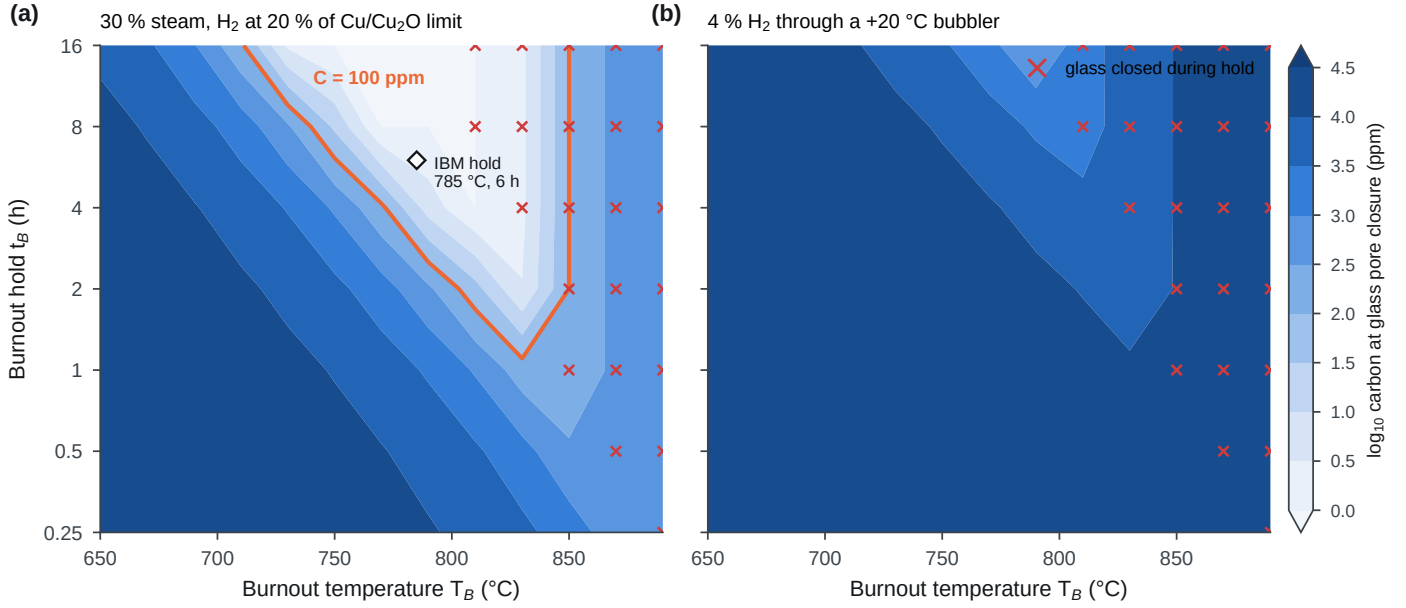


Figure 4: Carbon remaining in the glass-ceramic when its pores close, for burnout holds of temperature T_B and duration t_B within the baseline programme. (a) 30% steam with hydrogen at 20% of the Cu/Cu₂O limit; (b) 4% H₂ humidified at 20 °C. Orange: 100 ppm contour. Crosses: the glass sealed during the hold. Diamond: the IBM burnout hold [62].

temperature, when the line is pulled in tension by the rigid body around it while its own necks are still weak. The glass then catches up. Its shrinkage overtakes that of the copper at 889 °C, the bilayer bends the other way and reaches -259 m^{-1} (a radius of 3.9 mm) at 930 °C, and the final shrinkages differ by 4%: 26.3% for the glass-ceramic against 22.3% for the copper, both as logarithmic strains. For this material pair the difficulty lies less in the final shrinkage than in the order in which the two materials shrink.

4.6. Copper paste design maps

4.7. A matched paste and programme

4.8. The printed test vehicle

4.9. Cooling from the stress-free temperature

Below about 690 °C, the strain point of the calibrated glass, mismatch stresses no longer relax, and the expansion mismatch between copper ($16.5\text{--}22 \text{ ppm K}^{-1}$ over this range) and the glass-ceramic (3 ppm K^{-1}) is carried elastically. Fig. 7 shows the consequences for a copper plane on 1 mm of glass-ceramic. If copper remained elastic, its mean stress would exceed 900 MPa at room temperature and a 20 mm stack with 100 μm of copper would bow by 250 μm . Annealed copper cannot carry that stress. It yields within the first 10 K of cooling and then follows its yield stress, which rises from about 25 MPa at 690 °C to 69 MPa at room temperature. The bow falls accordingly, to 13 μm for 100 μm of copper and to 6 μm for 50 μm , and the glass-ceramic stays in mild compression (-7 MPa for 100 μm of copper). Plastic accommodation makes the thermal-expansion mismatch a secondary problem compared with the sintering mismatch, provided the copper is dense and ductile. The price is cyclic plasticity in the conductor, which matters for thermal cycling in service but is outside the scope of this study.

4.10. Which inputs decide the outcome

5. Discussion

6. Conclusions

CRediT authorship contribution statement

[Author names]: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The simulation code, input parameters with their provenance, and the scripts that generate every figure in this paper are available from the corresponding author on reasonable request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used [name of tool] in order to [reason, e.g. assist with code development, data analysis and drafting of the text]. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

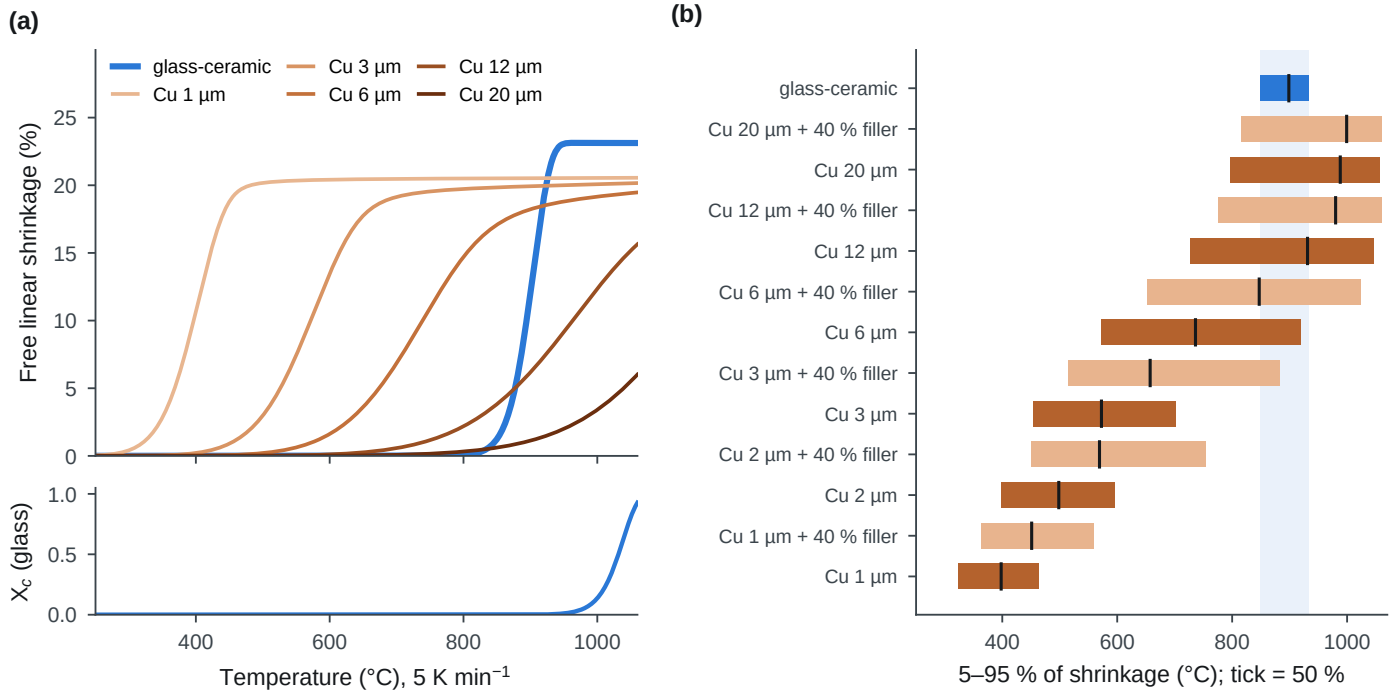


Figure 5: Free sintering of the two materials at 5 K min⁻¹ in dry 4% H₂ after binder removal. (a) Linear shrinkage of the glass-ceramic and of copper of different median particle size, and crystallised fraction of the glass. (b) Temperature range between 5 and 95% of the final shrinkage, with the midpoint marked, for copper with and without 40 vol% inert filler; the shaded band is the range of the glass-ceramic. Bars that reach 1060 °C had not finished densifying.

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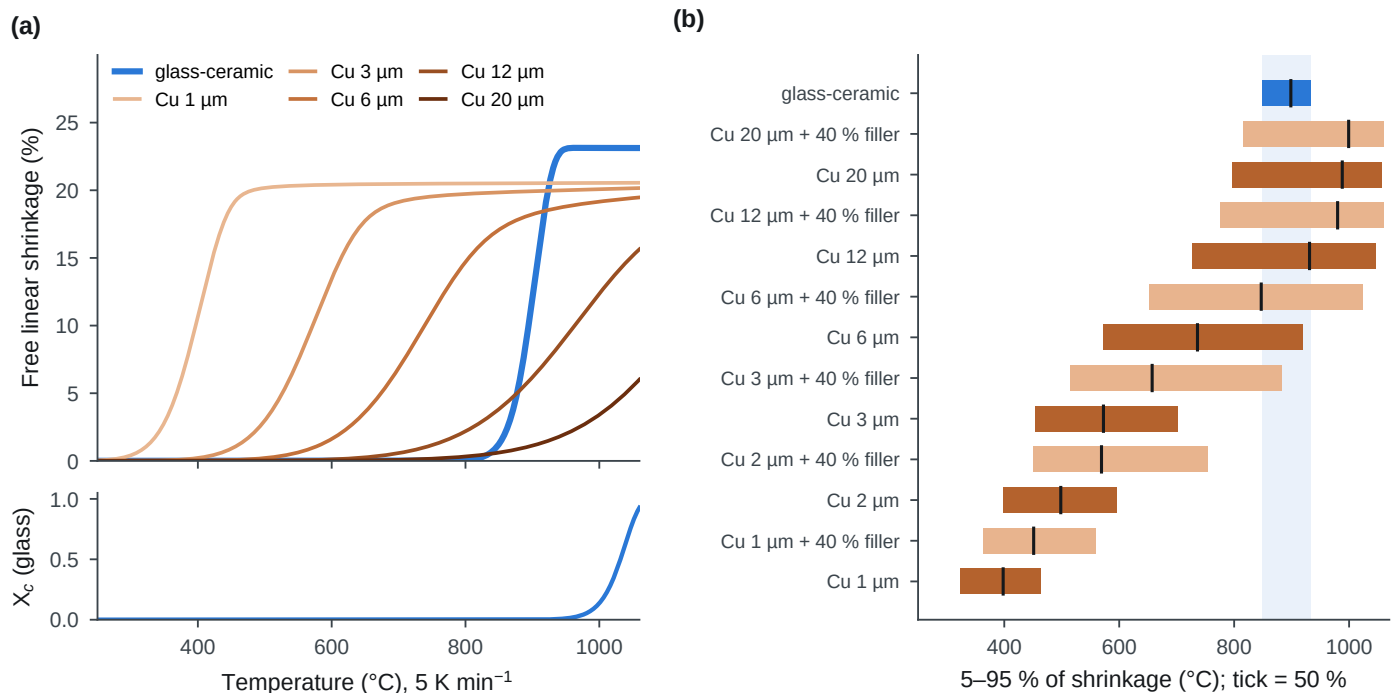


Figure 6: Co-firing histories. (a) Baseline: 3 μm copper without filler, 4 h steam burnout at 780 °C, 1 h at 960 °C. (b) Optimised paste and programme (Table ??). From top: furnace set point and part temperature, with the atmosphere of each segment shown by the band at the bottom (grey, N_2 ; green, steam; blue, dry H_2/N_2); relative densities and crystallised fraction of the glass; free logarithmic shrinkage of each material, the shaded area being the mismatch; carbon in the glass-ceramic and damage index of an embedded copper line.

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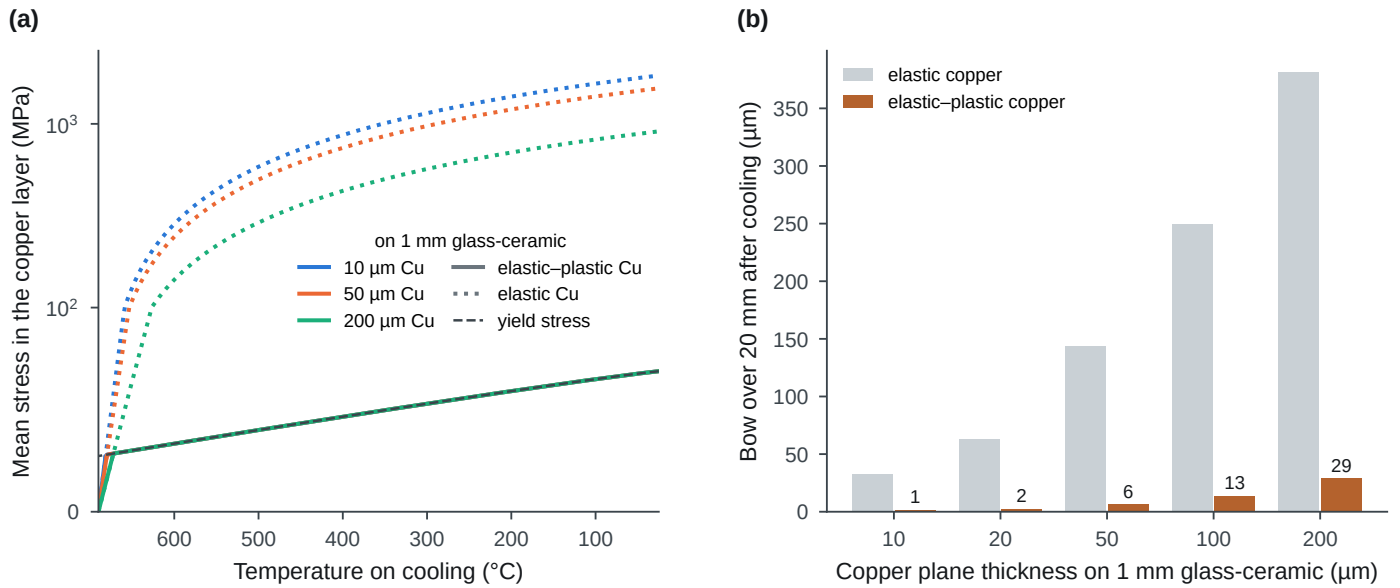


Figure 7: Cooling of a copper plane on 1 mm of glass-ceramic from the stress-free temperature (690 °C). (a) Mean stress in the copper layer for elastic and elastic-perfectly plastic copper; the plastic curves coincide with the temperature-dependent yield stress. (b) Bow over a 20 mm span at room temperature.

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Appendix A. Copper slab model and its calibration

The copper conductor uses the slab model developed for single-material copper parts printed from a photocurable slurry. Besides the binder chemistry of Section 2.2, it tracks the oxygen bound in the powder as Cu_2O . The oxide grows in oxygen-bearing gas by a decelerating law fitted to powder oxidation data, is reduced by hydrogen above the equilibrium ratio of Eq. (9), and reacts with char by $\text{Cu}_2\text{O} + \text{C} \rightarrow 2\text{Cu} + \text{CO}$. None of these oxidation paths is active in the co-firing programmes studied here, because the inlet gas contains no oxygen and the water-to-hydrogen ratio is held below the copper–oxide equilibrium. What matters for co-firing is the sintering viscosity. Eq. (7) uses the lattice and grain-boundary diffusion data for copper tabulated by Frost and Ashby [52], with one factor, f_η , fitted to the densification of copper parts printed from a 60 vol% photocurable slurry of 22 μm powder and sintered for 4 h at 1050 $^\circ\text{C}$ in hydrogen, which reached 90–94% relative density [31]. The fit gives $f_\eta = 0.28$, with 0.22–0.34 spanning the reported density range. The powder size enters Eq. (7) through the grain size, set initially to 0.4 times the median particle size, and through the sintering stress. Transferring f_η from 22 μm to the 3–12 μm powders considered here therefore rests on the diffusional scaling of Eq. (7); this is why f_η is included in the sensitivity analysis of Section 4.10.

Appendix B. Verification

Viscous densification of the glass-ceramic. With crystallisation and char removed, and open porosity, Eq. (3) with a Newtonian matrix gives $\dot{\theta} = -9\gamma\theta/(4r\eta)$. The slab model was run through isothermal holds at 810, 830 and 850 $^\circ\text{C}$ with 4, 8 and 16 cells. The simulated porosity follows the closed form, integrated along the simulated part temperature, to within 1.7×10^{-4} over the full range of densification before pore closure, independently of the mesh (Fig. 2a).

Bilayer curvature. The three-dimensional model of a free 12 $\times$ 12 mm plate made of 0.6 mm copper on 1.8 mm glass-ceramic was solved for the instantaneous velocity field at 900 $^\circ\text{C}$, and the curvature rate was fitted to the mid-plane deflection. It converges to the viscous Timoshenko value of Eq. (10) with second-order behaviour: the error falls from 3.4% at 0.6 mm voxels to 0.69, 0.22 and 0.05% at 0.3, 0.2 and 0.15 mm (Fig. 2b).

Free sintering and mass conservation. A free, frictionless cube held isothermally densifies homogeneously and isotropically, matches the closed-form Skorohod–Olevsky solution to 0.22% in density, and conserves solid mass to 0.07%.

Crystallisation kinetics. With the rate constant calibrated as described in Section 2.3, the simulated exotherm at 10 K min $^{-1}$ peaks within 0.01 K of the prescribed temperature for Avrami exponents between 1 and 4.

Table 1: Model inputs. Basis: L, literature value for this material; A, analogue material; C, calibrated in Section 3.2; D, derived; E, estimate. Ranges in the last column are those used in the Morris screening.

Quantity	Symbol	Value	Basis	Source, remark or screening range
<i>Glass-ceramic body</i>				
Median particle size	d_{50}	$3.0 \mu\text{m}$	A	IBM glass powders $2\text{--}7 \mu\text{m}$ [63]; $2\text{--}7 \mu\text{m}$
Size span $(d_{90} - d_{10})/d_{50}$	–	1.5	E	
Solids loading of the slurry	ϕ_{GC}	0.45	E	0.40–0.50
Density of the dense body	ρ_s	2600 kg m^{-3}	A	dense α -cordierite glass-ceramic, 2620 kg m^{-3} [64]
Surface energy	γ	0.36 J m^{-2}	L	calculated for a cordierite-type glass [65]; 0.24–0.36, since water vapour lowers the surface tension of silicate glass [66]
Glass transition (10^{12} Pa s)	T_g	$734 ^\circ\text{C}$	C	$714\text{--}754 ^\circ\text{C}$
Fragility index	m	32.4	C	28–40
High-temperature viscosity limit	$\log_{10} \eta_\infty$	–3	L	[48]
Crystallisation activation energy	E_c	303 kJ mol^{-1}	L	IBM-type high-cordierite glass [67]; $330\text{--}440 \text{ kJ mol}^{-1}$ in other $\text{MgO--Al}_2\text{O}_3\text{--SiO}_2$ glasses [68, 69]; $250\text{--}470 \text{ kJ mol}^{-1}$
Avrami exponent	n	2	E	1.5–3.7; 2.8–3.7 in a related $\text{MgO--Al}_2\text{O}_3\text{--SiO}_2$ glass [69]
Exotherm peak at 10 K min^{-1}	$T_{p,c}$	$1040 ^\circ\text{C}$	C	$1000\text{--}1080 ^\circ\text{C}$
Maximum crystal packing	X_m	0.64	E	random close packing
Density at pore closure	ρ_{cl}	0.92	E	
Thermal conductivity, dense	k	$2.8 \text{ W m}^{-1} \text{ K}^{-1}$	A	copper co-fired LTCC GL570 [70]
Young’s modulus, dense	E_{GC}	128 GPa	A	GL570 [70]
Poisson’s ratio	ν_{GC}	0.25	E	
Thermal expansion coefficient	α_{GC}	3.0 ppm K^{-1}	L	production glass-ceramic substrate [71]
Flexural strength	$\sigma_{s,GC}$	210 MPa	L	glass #12 fired in $\text{H}_2/\text{H}_2\text{O}$ [62]
Stress-free temperature	T_{sf}	$690 ^\circ\text{C}$	D	strain point ($10^{13.5} \text{ Pa s}$) of the calibrated glass
<i>Copper conductor</i>				
Median particle size	d_{50}	$1\text{--}20 \mu\text{m}$	–	design variable; baseline $3 \mu\text{m}$
Size span	–	1.2	E	
Solids loading	ϕ_{Cu}	0.40–0.60	–	design variable; baseline 0.50
Inert filler fraction	f	0–0.40	–	design variable; baseline 0
Viscosity factor	f_η	0.28	C	Appendix A, from [31]; 0.1–1.0
Lattice and boundary diffusion	$D_v, \delta D_b$	–	L	[52]
Carbon inhibition scale	C_{inh}	200 ppm	E	
Young’s modulus	E_{Cu}	117 GPa	L	oxygen-free copper [72]
Poisson’s ratio	ν_{Cu}	0.34	E	
Thermal expansion coefficient	$\alpha_{Cu}(T)$	$16.5\text{--}23.7 \text{ ppm K}^{-1}$	L	instantaneous values, 293–1100 K [73]
Yield stress	$\sigma_y(T)$	69 MPa at $20 ^\circ\text{C}$	L	annealed temper [72]; softening $1 - T^{*1.09}$ [74, 75]
<i>Binder, char and atmosphere</i>				
Char yield of network and backbone	χ	0.05	E	0.02–0.10
Gasification half-life, $800 ^\circ\text{C}$, reference gas	$t_{1/2}$	2 h	E	0.5–8 h
Gasification activation energy	E_g	200 kJ mol^{-1}	A	order of steam gasification of carbons [76]; $160\text{--}240 \text{ kJ mol}^{-1}$
Hydrogen inhibition constant	K_H	25	E	form after [77]
$\text{Cu}/\text{Cu}_2\text{O}$ and $\text{C}/\text{H}_2\text{O}$ equilibria	K_{Cu}, K_{st}	–	L	NIST-JANAF [46]
Safety factor on the $\text{Cu}/\text{Cu}_2\text{O}$ limit	s	0.2	E	
<i>Geometry</i>				
Slab half-thickness	–	$2.0 / 0.15 \text{ mm}$	E	glass-ceramic / copper
Reference bilayer	h_{Cu}, h_{GC}	0.1, 1.0 mm	E	20 mm span

Table 2: The IBM worked schedule [62] simulated at three hold temperatures. Shrinkage is the fraction of the total fired shrinkage reached by the end of the 6 h hold; carbon is the residual carbon in the glass-ceramic at the end of the hold.

Hold temperature	750 °C	785 °C	830 °C
Shrinkage during hold	7%	47%	100%
Pores at end of hold	open	open	closed
Carbon at end of hold (ppm)	33	0.006	0.07
Final relative density	0.990	0.990	0.990
Final crystallised fraction	1.00	1.00	1.00
Statement in [62]	too slow	chosen	pores close